

The Pilot Inquiry

The National Science Foundation asked whether the company would apply to a multi-million-dollar pilot program. This packet answers exactly: the pilot's published gate, where the company stands against it, and the Pitch that takes the first step today.

Day 4 of 5

Repository v4.9.0

3 figures

5 tables

Pitch first, pilot later

The pilot
\$20M

Six to eight awards of \$1M to \$3.75M

The gate
Phase II

An NSF award the company does not yet hold

The first step
\$305K

Phase I, opened by the Pitch submitted today

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Repository (v4.9.0) is at <https://github.com/kevinkawchak/physical-ai-oncology-trials>. The paper files are deposited in /funding/auto-fund-2. No digital object identifier is asserted for this document.

Keywords National Science Foundation; SBIR Phase I; Project Pitch; Commercialization Readiness Pilot; Strategic Breakthrough; Language model verification; Robotic surgery; Pancreatic ductal adenocarcinoma

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Abstract

The National Science Foundation has asked whether ChemicalQDevice would apply to a multi-million-dollar pilot program. In September 2026 the agency announced a \$20 million, two-year Commercialization Readiness Pilot, run by the National Commercialization and Translation Institute at the University of Central Florida, that will select six to eight companies for milestone-based awards of \$1 million to \$3.75 million each [1, 2]. Its competition is open to active or recent NSF SBIR and STTR Phase II and IIB awardees. The

agency's Strategic Breakthrough tier, of up to \$30 million with a one-to-one match, is likewise open only to Phase II awardees [3]. The company holds no NSF award, and so meets neither gate today.

This packet answers the inquiry exactly. It thanks the agency, states the gate and the company's place against it, asks whether the inquiry contemplates a route the company has missed, and submits the Project Pitch that is the first step on the published route: Pitch, Phase I, Phase II, and only then the pilot. NSF's SBIR program funds technical innovation and does not fund clinical trials, so the Pitch proposes the part of the program NSF can fund: a governed, advisory-only language model layer for robotic oncologic surgery, with verification by independent models and a public audit trail. The robotic Phase 1 itself stays with the National Institutes of Health and private capital, and the packet shows how the program is split so that no dollar is asked for twice.

How to read this

A reader with five minutes takes §1, the inquiry, the gate and the funding ladder, and §2, the day's actions with the path from Pitch to proposal.

A reviewer asking what NSF is being asked to fund takes §3, the split between NSF, NIH and private capital, and §4, the four Phase I objectives, each with a measure and a threshold.

A funder takes §5, the merit review criteria against the program's evidence, including the quantitative record behind the robotic Phase 1. §6 carries the positioning constraints, the method, and the references, every one of which is a clickable link.

1 The Inquiry, and the Gate in Front of the Pilot

1.1 What was asked, and what the answer must be

An inquiry to apply is an honor, and it is also a question with an exact answer. The pilot the inquiry concerns is new: NSF announced it in September 2026 as a way to carry deep technologies from its small business portfolio toward market readiness, with the Institute at the University of Central Florida selecting the companies and managing their milestones [1]. The gate to it is published, and the company does not yet pass it. The answer therefore does three things and no more: it thanks the agency, it states the gate and the company's place against it in writing, and it asks whether the inquiry had a different route in mind. The letter to the pilot's principal investigator asks the same question of the Institute [2].

1.2 The tiers and pilots, side by side

Table 16 sets every NSF tier and pilot the inquiry could concern beside its amount, its eligibility, and the company's position today. The amounts are those published in the current solicitations and the pilot's announcement [3, 4, 1]. Only one row is open to the company now, and it carries no money: the Project Pitch, which NSF requires before any Phase I proposal.

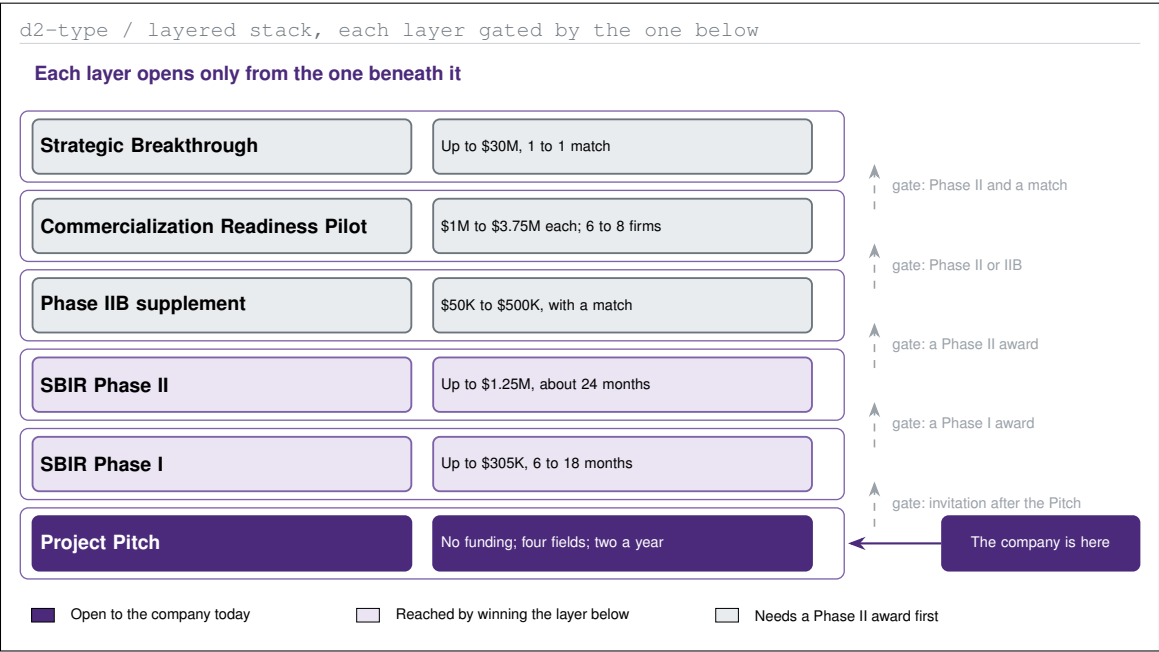


Figure 10. The NSF tiers and pilots as a layered stack, each opened only by an award or a match on the layer below, with the company on the Pitch layer at the bottom.

Tier or pilot	Amount	Who is eligible	The company
Project Pitch	None; four short fields	Any small business	Submitted today
SBIR Phase I	\$305K; 6 to 18 months	Invited after a Pitch	Once invited
SBIR Phase II	\$1.25M; about 24 months	Phase I awardees	After Phase I
Phase IIB supplement	\$50K to \$500K, with a match	Phase II awardees	Ledger at \$0
NCTI Readiness Pilot	\$1M to \$3.75M each, of \$20M	Recent Phase II or IIB	Not yet eligible
Strategic Breakthrough	Up to \$30M; 1 to 1 match	Phase II awardees	Not yet eligible
Instrumentation, 26-511	Share of a \$40M emphasis	Instrument proposers	Outside scope

Table 16. The NSF tiers and pilots the inquiry could concern, each with its amount, who is eligible, and the company’s place today; only the Project Pitch is open to it now.

1.3 The same tiers, drawn as a ladder

Figure 10 draws Table 16 as the stack it really is. Each layer is reached only through the one beneath it, and the gate between two layers is an award, a match, or an invitation. The company stands on the lowest layer. The pilot the inquiry concerns sits two gates above the first award the company could win.

1.4 The standing fact

The trial behind the program is built around daraxonrasib, which the FDA approved on August 26, 2026 as Rasonque for metastatic pancreatic adenocarcinoma after prior systemic therapy [5, 6, 7]. The approval followed Revolution Medicines’ Phase 1/2 program and its Phase 3 RASolute 302 trial [8, 9, 10]. ChemicalQDevice’s AI papers on the molecule were developed independently of that program and simultaneously with it, from the June 2025 identification study [11] to the Phase 1 protocol and investigational new drug application of 2026 [12, 13]. The perioperative use the company’s protocol proposes remains investigational, and a separate adjuvant Phase 3 trial of the molecule has been enrolling since December 2025 [14].

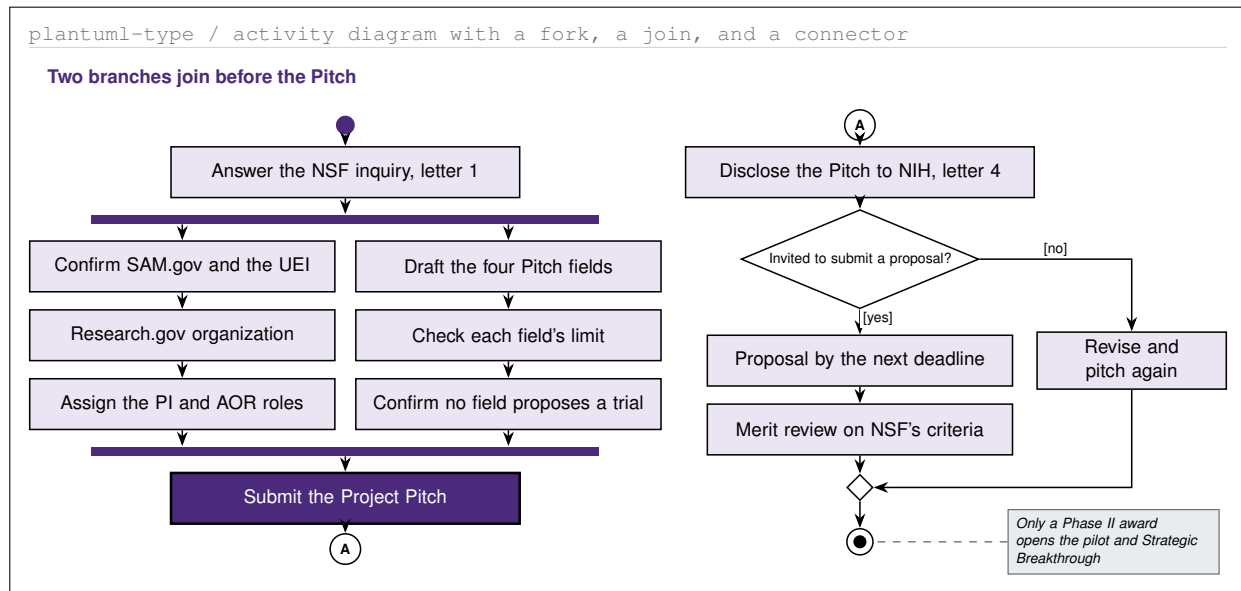


Figure 11. The route from the answer to a proposal, with registration and drafting run in parallel and joined before the Pitch, then one decision that NSF alone makes.

2 The Action Register

Table 17 lists the day's seven actions in the order they are done. The answer to the inquiry goes first, because every other letter refers to it. The Pitch waits on the organization record, and the letter to NIH waits on the Pitch, because it reports the Pitch as submitted.

No	Action	File	Waits on
1	Answer the NSF inquiry	email-01	Nothing
2	Ask the pilot's gate of its investigator	email-02	Row 1 sent
3	Confirm the Research.gov record	form-02	An active SAM.gov record
4	Submit the Project Pitch	form-01	Row 3
5	Ask about NAIRR Pilot resources	email-03	Nothing
6	Disclose the Pitch to NIH	email-04	Row 4 submitted
7	Open the match ledger	capital-04	Nothing

Table 17. The day's seven actions in the order they are done, each with the file that carries it and what it waits on; the letter to NIH follows the Pitch it discloses.

2.1 From the Pitch to a proposal

Figure 11 draws the route from the answer to a proposal as an activity diagram. Two branches run in parallel and must join before the Pitch is submitted: the registration branch, which confirms the company's federal entity record and its Research.gov organization [15], and the drafting branch, which writes the four Pitch fields and measures each against its character limit. The diagram breaks at a connector and resumes at the right. NSF then invites a full proposal or declines to [3]. A declined Pitch is revised and submitted again, and the solicitation allows two Pitches a year.

2.2 What each letter asks

Letter 1 thanks NSF's SBIR program for the inquiry, states the gate, and asks whether the inquiry contemplates a route the company has missed. It copies the Policy Office, which owns the rules the gate is written

in, and the Research.gov help desk, which would resolve any registration fault [16, 17]. Letter 2 asks the pilot's principal investigator at the University of Central Florida to confirm the gate in writing, and asks for nothing else [2]. Letter 3 asks the NAIRR Pilot whether a small business can request shared resources for verification research on synthetic data only [18, 19]. Letter 4 tells the NIH SBIR program office and the National Cancer Institute's SBIR team that an NSF Pitch has been submitted on the verification work, so that any later NIH application discloses it and neither agency pays for the same work twice [20, 21, 22].

2.3 A note on the date

October 1 is the first day of the federal fiscal year. If appropriations have lapsed, NSF staff may not answer for a time, and a portal may accept a submission it cannot yet process. The Pitch can still be submitted. Nothing else is followed up until three business days after the agency reopens.

The one approval this day asks for

Approve answering the NSF inquiry, testing the pilot's published eligibility gate in writing with its principal investigator, and submitting the Project Pitch that opens the only route to it. Approve disclosing the Pitch to NIH on the day it is submitted.

3 How the Program Is Split

3.1 The rule that shapes the split

NSF's SBIR program funds technical innovation with commercial potential, and it does not fund clinical trials [3]. The National Institutes of Health fund clinical research, including small business work that leads to a trial [20, 21]. The program's private capital sits behind a firewall that gives no investor any say in trial conduct, and pays for what neither agency pays for [23]. The split in Table 18 follows those three facts and nothing else. No component is marked yes in two public columns.

Component	NSF SBIR	NIH SBIR	Private capital
Verification method for the model	Yes, the Pitch	No	No
Boundary tests and audit trail	Yes	Cites the NSF result	No
Timing tests, hardware in the loop	Yes	No	No
Protocol, IND, site activation	No	Yes	In part
Drug, surgery, and follow-up	No	Yes, at the site	In part
Robot platform, site build-out	No	In part	Yes
Commercializing the method	After Phase II	No	Yes

Table 18. The program's seven components split between NSF, NIH and private capital, with no component funded twice; the verification work is the part proposed to NSF.

3.2 Why the scopes cannot overlap

The NSF work ends where a patient begins. It produces a verification method, a test suite and measured failure rates on synthetic cases generated from published parameters. The NIH work begins where a patient begins: it uses the verified layer inside a trial that belongs to a site investigator, under an investigational new drug application filed under 21 CFR Part 312 [24]. If both were funded, the NIH work would cite the NSF results rather than repeat them, and each application would disclose the other, as the SBIR policy directive requires [22]. Letter 4 tells NIH so in writing, with NSF copied.

3.3 The money on each side

An NSF Phase I is worth up to \$305,000 [3]. The company's plan carries an NIH Phase I of \$306,000 and an NIH Phase II of \$1,300,000, for an SBIR route of \$1,606,000, and places \$5,900,000 of private capital behind the firewall against the \$2,104,000 delta the route does not buy [23]. The NSF and NIH Phase I figures are nearly equal and pay for different work, and neither is counted twice anywhere in the company's documents. The private figure is a plan, not a completed raise, and the day's match ledger records its opening balance as zero.

3.4 What the split means for the trial

None of the robotic Phase 1 is proposed to NSF. What NSF would fund is the evidence that the advisory layer inside that trial stays inside its boundaries. That is the evidence an institutional review board and the FDA will ask for before the layer is allowed near a patient [25], and it is the evidence a clinical funder will ask for before paying for the trial it sits inside. The NSF work therefore does not compete with the NIH route. It is the step that makes the NIH route fundable.

4 The Phase I Technical Objectives

4.1 What the Pitch proposes

The Pitch proposes a governed, advisory-only language model layer for robotic oncologic surgery, and a verification method that makes its behavior checkable. The layer runs on premises at a clinical site. It produces text recommendations to the operating surgeon and nothing else: it holds no write credential to the data capture system, no route to the robot control network, and no ability to act on equipment [12, 13]. Every output is checked by two independent models from different developers before it is shown, a practice the company already follows in its own work [26], and every check is written to an audit trail that can be replayed.

The innovation is the combination of three things that clinical AI systems usually treat separately: a hard architectural boundary, enforced by network segmentation and credential design rather than by instructions to the model; cross-model verification, in which disagreement between independent models is a signal to withhold advice rather than noise; and a credibility assessment framed on the ASME V&V 40 approach to computational models [27], extended from simulation to advisory text.

4.2 Five measures, each with a threshold

Table 19 states the Phase I as five measures. The first four are the technical risks the Pitch names; the fifth restates the physical limits of the deposited protocol that the layer must never touch. Each row has a threshold a reviewer can check at the end of the award.

Objective	Measure	Threshold	Source
Boundary integrity	Open paths across the boundary, tested on every build	Zero write credentials; no route to robot control	Protocol, IND
Verification yield	How often verifiers disagree, and flag a seeded error	A withholding threshold set from measured rates	Company method
Credibility	An ASME V&V 40 battery, extended to advisory text	Twin at 81.9 over 55 tests; extension with limits	Deposited twin
Timing	Latency against each safety stop, on a hardware bench	No effect on 3 ms arm or 500 ms system stops	Protocol
Physical limits	Any advisory path to a force or vessel control	Zero paths to the 3 N, 18 N and five-vessel limits	Protocol

Table 19. The Phase I as five measures, each with a threshold a reviewer can check at the end of the award and the deposited source it comes from; no patient data is used.

4.3 What the Phase I would produce

A Phase I would produce three things: a verification method, the test suite that exercises it on every build, and a public report of measured rates with their limitations. The main technical risks are building synthetic cases realistic enough to expose failure, choosing verifier models that fail independently rather than together, and measuring rare events without patient data. Each is a question of engineering and measurement, which is the kind of question NSF funds.

4.4 What is not proposed

No clinical trial is proposed, and no patient data is used, identifiable or de-identified. The deployed clinical model stays on premises at the site, and any shared computing resource would see only synthetic cases and the company's own code. The robotic Phase 1 the layer would one day sit inside remains an open-label, single-arm, 3+3 dose escalation of perioperative daraxonrasib at 160, 220 and 300 mg, with a 28-day dose-limiting toxicity window, in up to 18 treated participants undergoing a staged eight-arm robotic pancreaticoduodenectomy with 56 degrees of freedom and 640 sensor channels [12]. That trial belongs to a site investigator and to NIH and private funding, as §3 sets out.

5 The Merit Review Map

NSF reviews every proposal against two criteria, Intellectual Merit and Broader Impacts, and judges SBIR proposals on their commercial impact as well [17, 3]. Table 20 maps each question a reviewer will ask to the evidence the company can put in front of that reviewer today, and names the evidence it does not yet have. The last column is as important as the third: a reviewer trusts a map that shows its own gaps.

Criterion	What NSF asks	The evidence today	Not yet in hand
Intellectual Merit	Advances knowledge?	Model disagreement used as a signal to withhold	Measured rates
Intellectual Merit	Well reasoned?	A hard boundary, not instructions; ASME V&V 40	An outside review
Intellectual Merit	Team qualified?	A dated public record since June 2025	Staff, on award
Intellectual Merit	Resources adequate?	On-premises compute; a NAIRR request	A NAIRR reply
Broader Impacts	Benefit to society	Models used only once shown to stay in bounds	Phase I evidence
Broader Impacts	Workforce	Eleven planned roles, six needing no license	A first hire
Commercial impact	Who pays, and why?	Centers, sponsors and robot makers, for a checkable boundary	A paying customer

Table 20. The seven questions an NSF reviewer asks under the three criteria, with the evidence the company can show today and the evidence it does not yet have in hand.

5.1 Where compute, data and review would sit

The resources question is answered by a drawing rather than a sentence. Figure 12 places every component in one of three clusters: the clinical site, where the model runs and where patient data would stay; the shared NAIRR Pilot resources, which would see only synthetic cases and the company's code if a request were granted [19]; and the public record, which receives the method and the measured rates. The robot control network is a cluster of its own inside the site, and the model has no route into it.

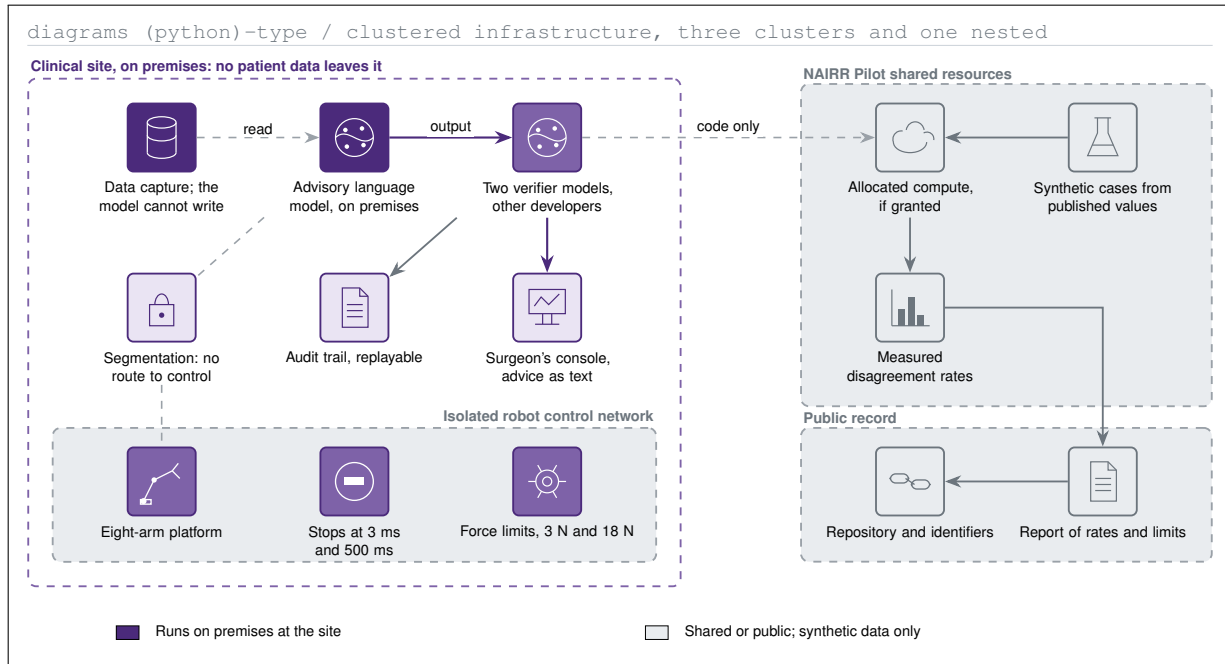


Figure 12. Where compute, data and review would sit, with the model and its verifiers on premises, the robot network sealed off, and only code and synthetic cases shared.

5.2 The quantitative record behind the robotic Phase 1

A funder of the verification work is also, one step removed, a funder of the trial it would make possible, and that trial rests on a dated quantitative record. In August 2025 the company's ten-arm simulation of the agent returned a median overall survival of 12.8 months against 5.4 for chemotherapy [28]. In May 2026 the developer's Phase 3 trial reported 13.2 months against 6.6 in the RAS G12 population of previously treated metastatic disease [10], and the approval that followed reported 13.2 against 6.7 months in the overall population [5]. The company describes the closeness of the simulated and reported figures as a chronology, not a validation, because the populations, regimens and sample sizes differ. Pancreatic ductal adenocarcinoma remains the cancer with the widest gap between incidence and survival in the United States [29], and after resection the standard remains combination chemotherapy [30].

The program's direct cost is \$700,000 a year for five years, of which \$521,000 is personnel across 3.95 full-time equivalents [31]. The company's comparable virtual trial work is **projected** at \$36,330 per run and about one month, against industry benchmarks above \$120,000 and 4.5 months [23]. That figure describes simulation work, not the cost of the clinical trial.

6 Positioning, Method, and References

6.1 Positioning constraints

An inquiry to apply is not an invitation to submit a proposal, an award, or an assessment of eligibility, and nothing in this packet describes it as one. The company holds no NSF award, and it is not eligible today for the Commercialization Readiness Pilot or the Strategic Breakthrough tier [1, 3]. No agreement of any kind exists with the National Science Foundation, the University of Central Florida, the NAIRR Pilot, or the National Institutes of Health, and none is described as a partner, sponsor, or endorser. The Pitch does not ask NSF to fund a clinical trial, and no patient data is used in the work it proposes.

Rasonque is approved for metastatic pancreatic adenocarcinoma after prior systemic therapy [5], and the perioperative use this program proposes remains investigational. No investigational new drug application has

been submitted, and no patient has been treated. The private capital figure is a plan, not a completed raise, and nothing in the day's capital instruction is investment advice.

6.2 Method and reproducibility

This packet was produced from repository sources and from the public pages of the offices it writes to: the tiers, amounts and gates from the current NSF solicitations and the pilot's announcement [3, 4, 1]; the split and the money frame from the capitalization plan [23]; the technical objectives from the deposited protocol and investigational new drug application [12, 13]; the verification practice from the company's three-model method [26]; and the staffing figures from the site package [31]. Every recipient address was checked against the page its office publishes, and every Pitch field was measured against its character limit before it was committed.

The three figures are drawn in TikZ from the specifications in the day's diagrams directory, one each in the D2, PlantUML and Diagrams vocabularies, and no raster image is used anywhere. The human author retains responsibility for every conclusion, and this document was adapted using Claude Code Opus 5.5 [32].

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